

DLMI FINAL PROJECT

The final project requires students to develop and evaluate a deep-learning solution to a relevant scientific problem in medical imaging. Suitable topics include (but are not limited to) semantic image segmentation, object detection, image registration, image reconstruction, artifact correction, and image synthesis.

A list of instructor-prepared options is provided below. Students may choose one of these options or propose their own. Custom proposals are subject to instructor approval based on the suitability of the problem, the level of technical complexity, and alignment with the course material. Projects may be completed individually or in teams of two. Multiple teams may work on the same project, provided that each team develops and evaluates its own independent solution and does not share code, results, or implementation.

Project deliverables include a written report and a fully functional software solution, preferably implemented in Python using Jupyter notebooks and PyTorch. The report should clearly document the software environment, dataset(s), mathematical and algorithmic methods, and quantitative evaluation of the results. Students are encouraged to follow established reporting guidelines for AI/ML studies in medical imaging, such as the references listed below:

- <https://pubs.rsna.org/doi/10.1148/ryai.240300>
- <https://aapm.onlinelibrary.wiley.com/doi/10.1002/mp.15170>

Deliverables

Software solution

1. Complete collection of Python files (.py) used in the solution.
2. Jupyter notebook files (.ipynb) used to run the solution.
3. Structured dataset(s).
4. Trained parameters for the instructors to reproduce your results.

Written report

1. Introduction
 - a. The purpose of the project and the rationale for using an AI/ML approach.
 - b. The expected contribution of AI/ML to the medical imaging application.
2. Materials
 - a. Characteristics of the dataset, including sample size and clinical acquisition sites (if known).
 - b. For phantom or simulated data: the type of phantom or simulation, a thorough description of the data-generation process, and a discussion of how well the simulated data represent realistic conditions.
 - c. The imaging modality or data type, including the acquisition or simulation protocol and relevant parameter ranges.
 - d. The suitability of the dataset composition for the intended AI/ML application.
 - e. A description of the ground truth or reference standard, including the annotation process, the degree of subjectivity involved, and any associated uncertainty.
 - f. The partitioning of the data into training, validation, and test sets, including any steps taken to reduce bias and the rationale for the chosen sample sizes.

3. Methods

- a. A sufficiently detailed description of the methodology to allow replication, including the model architecture, hyperparameters, inputs, input dimensionality (e.g., 2D or 3D), preprocessing steps, output definitions, and any discretization or binning procedures.
- b. A description of the training procedure, including the loss function, regularization strategy, methods for addressing class imbalance if needed, measures taken to reduce overfitting and bias, and any ablation studies performed.

4. Results

- a. The performance metrics used, including any post-processing steps applied to the AI/ML outputs, such as scoring criteria, decision thresholds, or binning.
- b. The method(s) used to estimate uncertainty in the reported performance metrics (e.g., 95% confidence intervals).
- c. The significance of the results relative to an appropriate null hypothesis, if applicable, or relative to a suitable benchmark.
- d. Representative results from the training, validation, and test sets.

Grading criteria:

Approach (35 points): Quality of the proposed method and evaluation approach. We'll consider whether the technical approach is appropriate for the problem, whether key design choices are well motivated, and whether the work goes beyond a minimal baseline. Strong projects typically include a clear problem formulation, a sensible model and training strategy, and some degree of thoughtful extension, comparison, or adaptation beyond what was covered directly in class.

Assessment (35 points): How well the method is tested and analyzed. We'll consider whether the evaluation uses appropriate metrics, whether results are reported clearly and quantitatively, and whether conclusions are supported by evidence. Strong projects typically include both quantitative and qualitative results, meaningful comparisons or baselines, and some discussion of limitations, uncertainty, or failure cases. Claims of improvement should be supported by the reported results.

Report (30 points): Clarity, completeness, and accuracy of the written report. We'll consider whether the report is well organized, whether the dataset, methods, and evaluation are described in enough detail to understand and reproduce the work, and whether the writing follows the reporting expectations for the project. Strong reports are precise, thorough, and honest about both strengths and limitations. Unsupported or inaccurate statements will be penalized.

#1 Fluoroscopic Guidewire Localization and Pose Regression

Background

Your goal is to estimate the tip location and angular orientation of guidewires in x-ray fluoroscopic images. In orthopaedic procedures, guidewires are used to guide the placement of larger instruments such as cannulated screws, and accurate placement often requires repeated imaging and manual adjustment. A reliable deep learning method for guidewire localization offers to reduce trial-and-error during the procedure and help quantify changes in position and orientation.

Approach

A general approach would estimate (e.g., regress) the guidewire tip location and orientation from a fluoroscopic image, ideally by predicting the 2D (x and y) coordinates of the tip together with its angular orientation (θ). To simplify the task, students may first crop or otherwise isolate a region containing a single guidewire and treat the problem as a regression task. Performance should be evaluated quantitatively using the Euclidean distance between predicted and true tip locations and the angular deviation from the true guidewire orientation.

Possible extension: A natural next step would be to combine the pose estimator with an object detection model that automatically identifies the guidewire bounding box.

Dataset

The dataset is provided as a single .npz file containing fluoroscopic images together with ground-truth annotations for guidewire position and direction. The images were acquired from multiple cadaveric specimens at several anatomical sites, including the pelvis, lumbar spine, thorax, and shoulders, using a mobile C-arm system. The dataset includes 314 images of size 976×976 , and each image contains exactly two guidewires; the annotations provide guidewire positions as (x, y) coordinates and directions as $(\Delta x, \Delta y)$.

#2 Segmentation of the Distal Tibiofibular Joint in CBCT

Background

Your goal is to segment the individual bones of the ankle joint – namely, the fibula, tibia, and talus. Accurate reduction and fixation of the ankle joint are important predictors of clinical outcome, and even small fibular displacements have been associated with post-traumatic osteoarthritis, pain, and disability. Automated segmentation of these bones from intraoperatively acquired cone-beam CT (CBCT) images would support quantitative assessment of joint reduction and tibiofibular spacing.

Approach

A practical starting point is to treat the problem as a multi-class segmentation task on 2D slices extracted from the 3D volumes. Students may work with axial, sagittal, or coronal slices and can improve performance through augmentation, including deformable transformations, and by resampling the volumes to create thicker slices (i.e., slabs) that provide additional anatomical context while retaining a 2D formulation. Performance should be reported separately for each bone using standard segmentation metrics.

Possible extension: A logical next step would be to move to a 3D solution using slice stacks or patch-based 3D inputs. For patch-based processing, check out the TorchIO library.

Dataset

The dataset consists of multiple .npz files, each containing one 3D CBCT image and its ground-truth segmentation labels for the fibula, tibia, and talus. In total, the dataset includes 40 ankle CBCT scans acquired on an OnSight 3D CBCT system. Each scan has dimensions of approximately $384 \times 384 \times 576$ voxels, with isotropic voxel spacing of $0.5 \times 0.5 \times 0.5 \text{ mm}^3$.

#3 CT-to-US Image Translation Using Paired Slices

Background

This project investigates cross-modal image synthesis, specifically generating ultrasound-like images from CT slices. The motivation is that deep learning methods in ultrasound often require large amounts of annotated data, yet ultrasound acquisition is highly variable across operators and acquisition settings. A method that can synthesize realistic ultrasound from CT could help generate larger training datasets and support downstream ultrasound analysis tasks.

Approach

A reasonable formulation is an image-to-image translation problem in which a model receives a CT slice and predicts the corresponding ultrasound image. Because the data are sequentially acquired while the ultrasound probe moves continuously along the anatomy, students may also exploit local temporal consistency across neighboring frames. Evaluation should emphasize both image quality and anatomical fidelity, including structural similarity and the avoidance of hallucinated structures that are not present in the target ultrasound image.

Possible extension: Possible extensions include uncertainty estimation and the more difficult reverse task of synthesizing CT from ultrasound.

Dataset

The dataset consists of paired CT and ultrasound images stored in .png format. The data are organized into three subject-specific subdirectories, and each subject folder contains multiple scans acquired during continuous probe motion. The recommended split is to use the first two subjects for training and the final subject, identified as subject #23-1831, for testing.

#4 Projection-Domain Segmentation of Catheters and Vasculature

Background

This project focuses on segmentation of catheters and vascular anatomy directly in cone-beam CT (CBCT) projection images. The data were synthetically generated to mimic interventional radiology procedures, specifically transarterial chemoembolization (TACE), in which a catheter is navigated through the arterial system to treat liver tumors. During these procedures, contrast-enhanced imaging is used to visualize the

downstream vasculature, and automatic segmentation could help identify the vessels supplying the tumor more efficiently.

Approach

Students may formulate the task either as segmentation of individual CBCT projections or as a multi-view learning problem that uses the full set of projections acquired for each scan. The goal is to segment the catheter, the vasculature, and the combined foreground regions, and performance should be reported for each of these targets. A strong baseline would be a 2D segmentation network applied independently to each projection, while more advanced approaches could exploit shared structure across views.

Possible extension: An initial approach may treat the vasculature and catheter as a single foreground class, without distinguishing between them, and later extend the method to segment them separately.

Dataset

The dataset contains synthetic CBCT projection images generated from 54 MDCT scans from the public CT-ORG dataset. For each base scan, artificial catheters and vasculature were inserted with attenuation values chosen to mimic intravenous contrast, and forward projections were then generated so that both structures are visible in the images. Each CBCT dataset includes 20 projection views sampled uniformly from -100° to $+100^\circ$ along the acquisition trajectory (with 0° oriented in the AP direction). In total, the dataset includes 1512 image-label pairs (28 different sets of catheters/vasculature for each of the 54 scans), and the provided `data_splits.npz` file defines train, validation, and test splits of approximately 70%, 20%, and 10%, respectively; importantly, splitting should be performed by base CT volume to avoid data leakage.

#5 Image Restoration for Motion-Corrupted Interventional CBCT

Background

Your goal is to implement motion artifact reduction for interventional cone-beam CT (CBCT) images. The clinical motivation comes from image-guided interventional radiology procedures such as transarterial chemoembolization (TACE), in which intra-procedural CBCT is used to localize target anatomy, visualize the vascular tree, and assess treatment delivery. In practice, residual respiratory motion, cardiac motion, and peristaltic motion can degrade these scans, producing streaks, intensity non-uniformity, blur, and anatomical distortion.

Approach

The task is to develop a deep learning method that takes a motion-corrupted 2D axial CBCT slice as input and predicts a corrected image with reduced artifacts and restored anatomy. Students may treat the problem either as a supervised image restoration task using paired corrupted and motion-free images, or as a conditional generative problem using unsupervised or semi-supervised methods. Evaluation should go beyond simple intensity-based criteria and include measures of residual artifact severity, preservation of anatomical boundaries, structural similarity, and avoidance of hallucinated structures that are absent from the motion-free reference image.

Dataset

The dataset consists of synthetic 2D axial CBCT slices derived from 88 MDCT scans in the public TCIA Lymph Node Abdomen dataset. For each case, a smooth deformable motion field was simulated with random amplitude between 0 and 30 mm, random frequency between 0.22 and 0.45 Hz, random phase, and a random location within the abdominal soft tissues; the axial slice location was also selected so that part of the liver was present. Each case includes a motion-corrupted image and a motion-free counterpart generated using high-fidelity forward projection and conventional FBP-like reconstruction. The reconstructed images are sampled on a 600×600 grid with $0.5 \times 0.5 \times 0.5 \text{ mm}^3$ voxel size.

#6 MR Angiography Segmentation of Whole-Brain Vasculature

Background

This project focuses on segmentation of the intracranial vasculature from magnetic resonance angiography (MRA). Accurate delineation of the cerebral vessel tree is important for characterizing vascular anatomy and could support downstream tasks such as localization of stenosis or occlusion, aneurysm analysis, report generation, and treatment planning.

Approach

The approach would be quite similar to that of Project #4, but in 3D. A reasonable initial approach is to formulate the problem as 3D vessel segmentation from a single-modality MRA volume, e.g., using a 3D U-Net or a variant. Evaluation should emphasize both overlap accuracy and preservation of vessel topology. Relevant metrics include Dice, class-average HD95, and cIDice.

Possible extension: An initial approach may treat all vasculature as a single label and focus on whole-vessel segmentation. A possible extension would then aim to distinguish individual vessels, either directly from the input images or by further processing the output of the initial vessel segmentation model.

Dataset

The dataset is derived from the 2025 MICCAI TopBrain challenge. MR images and corresponding segmentation labels are provided in .npy format. All images have been resampled to an isotropic voxel spacing of $0.5 \times 0.5 \times 0.5 \text{ mm}^3$. The definitions of the individual vessel labels are listed in a separate text file.

#7 Segmentation of Knee Bone and Cartilage in MRI

Background

Your goal is to segment the major osseous and cartilaginous structures of the knee from magnetic resonance (MR) images, specifically the femur, tibia, femoral cartilage, and tibial cartilage. Accurate segmentation of these structures is important for quantitative assessment of joint morphology and may support downstream analysis of cartilage loss, joint-space changes, and osteoarthritis-related degeneration. Because knee cartilage is thin and boundaries between adjacent tissues can be subtle, this is a meaningful and challenging medical image segmentation task. The SKI10 dataset was created specifically to benchmark algorithms for knee bone and cartilage segmentation from MRI.

Approach

The approach would be quite similar to that of Project #2. A practical starting point is to formulate the problem as a multi-class segmentation task on 2D sagittal slices extracted from the 3D MR volumes. Students may also explore 2.5D inputs, such as small stacks of neighboring slices, to provide additional anatomical context while keeping the model relatively simple. Standard augmentation methods such as flips, elastic deformations, intensity perturbations, and random cropping may be useful. Performance should be reported separately for bone and cartilage structures using standard segmentation metrics such as Dice and surface-distance measures.

Possible extension: A logical next step would be to move to a 3D solution using volumetric or patch-based inputs. Another possible extension would be to focus specifically on cartilage segmentation in anatomically relevant subregions, since cartilage remains the more difficult structure to segment accurately in this dataset.

Dataset

The dataset is derived from the SKI10 grand challenge on knee MRI segmentation. It contains 100 3D knee MR images with corresponding segmentation labels, and a commonly used split is 60 images for training and 40 for testing. The target labels are background, femur, femoral cartilage, tibia, and tibial cartilage. The images were acquired in the sagittal plane, with approximately 0.4×0.4 mm in-plane spacing and 1 mm slice thickness.